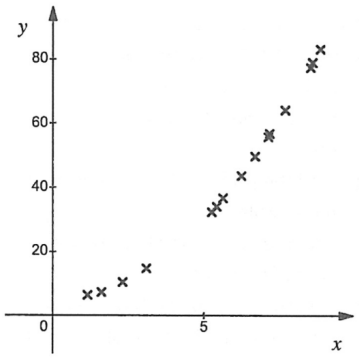
Bivariate Data: Association and Spearman's Rank

Questions 1–5

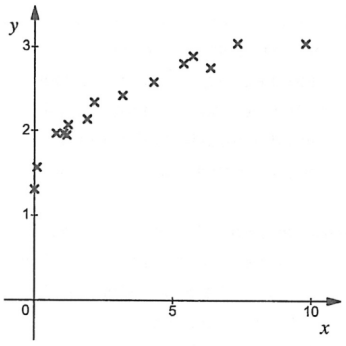
Question 1. The diagram below shows bivariate data.



Which of the following best describes the data shown in the diagram?

Positive correlation	Negative correlation
Positive nonlinear association	Negative nonlinear association

Question 2. The diagram below shows bivariate data.



Which of the following best describes the data shown in the diagram?

Positive correlation	Negative correlation
Positive nonlinear association	Negative nonlinear association

Question 3. Barnaby collects the bivariate data shown in the table below.

x	0.8	1.2	2.6	2.9	1.9	1.6	2.5	2.1
y	2.5	1.9	5.8	8.4	4.3	0.9	5.9	4.0

He ranks the data from smallest to largest and correctly finds the differences, d, and the squares of the differences, d^2, as shown in the table below.

Rank x	1	2	7	8	4	3	6	5
Rank y	3	2	6	8	5	1	7	4
d	-2	0	1	0	-1	2	-1	1
d^2	4	0	1	0	1	4	1	1

a) Given that $\sum d_i^2$ is the sum of the squares of the differences, and n is the number of pairs of bivariate data in the original table, calculate Spearman's Rank correlation coefficient using the formula

$$r_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$$

b) Which of the following best describes the association found in a)?

Strong Positive	Moderate Positive
Weak Positive	Weak Negative
Moderate Negative	Strong Negative

Question 4. Calculate Spearman's Rank correlation coefficient for the following bivariate data:

x	4.3	3.8	8.6	8.0	1.1	5.5	4.8	6.5
y	7.9	8.1	7.3	7.2	9.3	7.7	7.8	7.6

Question 5. Uzair wants to investigate whether there is an association between the number of hours students study, x, and their scores on a maths test, y. The data collected from a random sample of 8 students is shown in the table below.

Student	A	B	C	D	E	F	G	H
x	2	3	4	5	1	6	3	2
y	49	58	62	60	59	71	46	66

Calculate Spearman's Rank correlation coefficient for the data.